UNIT - IV

Handwritten Study Notes | JNTU(A) R23

Syllabus Topics:

Wetland hydrologic assessment: physical and biological processes | Anthropogenic and climate change impacts on wetland hydrology | Modeling wetland hydrology, hydraulics and hydrodynamics | Introduction to wetland treatment systems design | Water harvesting: rainwater, catchment harvesting, harvesting structures | Model watershed | Government and NGO projects

1. Wetland Hydrologic Assessment

Hydrologic assessment: collecting and analysing data on water sources, flows, levels and storage to understand how a wetland functions and how it responds to change. It is the first step in wetland conservation, restoration and design.

Steps of assessment

1. Delineate the wetland and its contributing catchment (topographic map, DEM, GIS).
2. Collect climate data: rainfall, temperature, solar radiation, wind, humidity -> ET.
3. Measure surface inflow and outflow (gauges, weirs, flumes, current meter), water levels (staff gauge, data loggers), tidal records.
4. Measure groundwater: observation wells, piezometers, seepage meters.
5. Prepare water budget, compute hydroperiod and residence time, check closure and uncertainty.
6. Link hydrology to soils (redox features, mottles), vegetation (indicator species) and fauna.

7. Use remote sensing (NDWI, SAR, Landsat/Sentinel) to map extent and seasonal change of water spread.

Physical processes

* Precipitation, runoff, infiltration, ET, groundwater exchange, tidal and wind effects, sedimentation and erosion, thermal stratification, evaporation losses.
* Sediment deposition changes depth and storage; erosion of banks changes channels; flow patterns change with vegetation drag.
* Wind-induced seiche and mixing in open-water zones; ice cover in cold regions.

Biological processes

* Vegetation: transpiration and shading modify ET and temperature, plant growth changes roughness (flow resistance) and traps sediment, peat accumulation raises the bed over time.
* Microbes: decomposition, nitrification-denitrification, sulphate reduction -> control water chemistry and carbon storage.
* Animals: beaver dams, crayfish burrows, fish and birds change hydrology and nutrient movement, invasive plants like water hyacinth increase ET and block flow.

> Eco-hydrology:
> Hydrology (physical) and ecology (biological) are coupled: hydroperiod shapes biology and biology feeds back on hydrology (eco-hydrology).

2. Anthropogenic and Climate Change Impacts

Human (anthropogenic) impacts

Activity	Hydrological impact
Drainage and reclamation	lower water table, loss of wetland area, reduced storage and flood buffer
Dams, barrages, diversions	altered flow regime, loss of floods/pulses, salinity change (Loktak, Keoladeo)
Urbanisation	impervious surface -> high runoff, polluted storm water, shrinking wetlands
Agriculture	nutrient/pesticide load, irrigation withdrawals, drainage ditches
Groundwater over-pumping	reduces seepage to wetland, springs dry
Deforestation in catchment	more erosion and siltation, flashy flows
Sewage and industry	pollution, eutrophication, loss of DO
Invasive species / over-fishing	choke channels, change ET and food web

Climate change impacts

* Temperature rise: higher ET, more evaporation loss, shorter hydroperiod, warmer water -> lower DO, algal blooms.
* Rainfall changes: more extreme storms and longer dry spells, shifts in monsoon timing, more floods and droughts.
* Sea-level rise: salt-water intrusion in coastal wetlands, loss of mangroves/marshes unless they can migrate inland.
* Glacier and snow-melt change affects Himalayan wetlands (Wular, Pangong, Tso Moriri).
* Feedback: wetlands emit methane (CH_4) and store carbon (peat, blue carbon), drying releases CO_2.
* Adaptation: restore wetlands as natural flood buffers, maintain environmental flows, protect buffer zones, integrated river-basin management, monitoring.

> Note:
> Impacts are often cumulative: human use + climate change together push wetlands over thresholds, so management must be adaptive.

3. Modeling Wetland Hydrology, Hydraulics, Hydrodynamics

Models are mathematical tools to simulate water level, flow and transport in a wetland for design, forecasting and impact assessment.

A. Hydrologic models (water budget)

* Lumped water-balance model: $dS/dt = P + Q_{in} - ET - Q_{out} +/- G$. Solve with time step (daily). Needs rainfall, ET, stage-storage and stage-discharge curves.
* Rainfall-runoff in catchment: SCS-CN, unit hydrograph, HEC-HMS, SWAT, VIC.
* Stage-storage curve S(h) from bathymetry/DEM; outflow rating curve Q(h) of weir/culvert.

$$S(t+dt) = S(t) + (P + Q_{in} - ET - Q_{out})\,dt$$

This reservoir-routing (level-pool or Modified Puls) approach gives the stage hydrograph.

B. Hydraulic models (flow in channels and structures)

* 1-D Saint-Venant equations (continuity + momentum) for unsteady open-channel flow:

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = q_{lat}\,; \quad \frac{\partial Q}{\partial t} + \frac{\partial (Q^2/A)}{\partial x} + gA\frac{\partial h}{\partial x} + gA\,S_f = 0$$

* Tools: HEC-RAS (1-D, 2-D), MIKE 11, SWMM; steady gradually varied flow by standard step method.
* Vegetated flow: roughness from stem density, drag coefficient, n depends on depth.

C. Hydrodynamic models (2-D / 3-D)

* Depth-averaged shallow water equations in 2-D (wetland with wide shallow areas); 3-D for stratified lakes and estuarine tidal marshes.
* Tools: MIKE 21, Delft3D, TELEMAC, EFDC, SFWMM (Everglades), MODFLOW (coupled groundwater), WASP and CE-QUAL for water quality; WETLAND-DNDC for biogeochemistry.
* Process: (1) conceptual model, (2) choose software, (3) grid and boundary conditions (inflow, tide, wind), (4) calibration (adjust n, K), (5) validation with independent data, (6) sensitivity and uncertainty, (7) scenario simulation (climate, management).
* Performance statistics: Nash-Sutcliffe efficiency NSE, RMSE, R^2, percent bias.

> Selecting a model:
> Model choice depends on purpose and data: lumped budget for water level, 1-D for channel, 2-D for sheet flow and mixing, coupled models for ground-surface exchange.

4. Introduction to Wetland Treatment Systems Design

Constructed (treatment) wetlands are engineered systems that use natural processes of wetland vegetation, soils and microbes to treat waste water, storm water, agricultural drainage, mine drainage.

Types

* Surface flow (FWS) wetlands: open water with emergent plants.
* Horizontal subsurface flow (HSSF): water below gravel bed.
* Vertical flow (VF): intermittent dosing from top, good aeration, nitrification.
* Hybrid systems (VF + HSSF) for N removal; floating treatment wetlands, reed beds for sludge drying.

Design steps

1. Define objective: pollutant, inflow Q, influent and target effluent quality (Cin, Cout).
2. Pre-treatment: screens, septic tank/settler to remove solids and grease.
3. Select type, site (flat, away from houses, soil permeability), climate.
4. Size the area: $A = Q \ln(Cin/Cout) / K$ (plug flow) or P-K-C* model for N tanks.
5. Depth (0.3 - 0.6 m), L: W ratio, bed slope, number of cells in series/parallel, inlet and outlet structures.
6. Media: gravel size 10 - 20 mm for SSF; liner (clay or HDPE) to protect groundwater.
7. Vegetation: Typha (cattail), Phragmites (reed), Scirpus, Canna, Colocasia, Vetiver; plant at 4 - 9 per sq m.
8. O and M: harvesting, sludge removal, water level control, mosquito control, monitoring.

Parameter	Typical design value
Hydraulic loading rate	0.02 - 0.1 m/day (SSF), 0.01 - 0.05 (FWS)
Organic loading	< 80 Kg BOD/ha/day (FWS), < 100 (SSF)
Area per person (domestic)	2 - 5 sq m (SSF) ; 5 - 10 (FWS)
Retention time	3 - 14 days
Removal BOD / TSS / N / P	70-90 % / 70-90 % / 30-60 % / 30-60 %

Mechanisms:

Treatment wetland mechanisms: sedimentation, filtration, adsorption, microbial degradation, plant uptake, nitrification-denitrification, UV and die-off.

5. Water Harvesting

Water harvesting: collection, storage and use of rain water (runoff) from rooftops, land surface or rock catchments for beneficial use or to recharge groundwater - 'catch the rain where it falls'.

A. Rainwater harvesting (RWH)

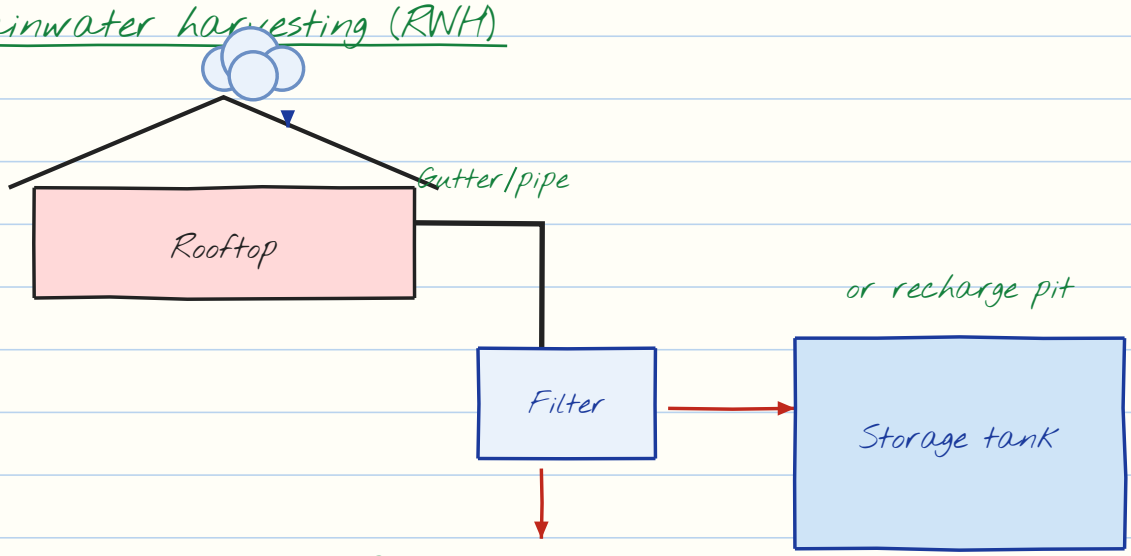


Fig 1: Rooftop rain water harvesting

* Components: catchment (roof), gutters and down pipes, first-flush diverter (discards the first 1-2 mm of rain), filter (sand, gravel, charcoal), storage tank (ferro-cement, RCC, plastic) or recharge structure.
* Recharge structures: recharge pit, trench, recharge well/borewell, injection well, percolation tank.

$$\text{Harvest } V = \text{Rainfall (m)} \times \text{Roof area (m2)} \times \text{runoff coefficient}$$

(C = 0.8 roof)

Example: 100 sq m roof, 800 mm rain, C = 0.8 -> V = 0.8 x 100 x 0.8 = 64 m3 = 64,000 litres per year.

* Tank size = daily demand x dry period days, maintain water quality by cleaning and chlorination.
* Advantages: free source, reduces storm water, recharges groundwater, improves quality (low TDS), cheap and decentralised. Mandatory for buildings in many cities (Chennai, Bengaluru, Hyderabad).

B. Catchment (runoff) harvesting

* Collect runoff from natural/treated land catchments into storage: ponds, tanks, small reservoirs.
* Catchment treatment: compaction, clearing, plastic/butyl lining, bitumen, concrete, chemical (sodium carbonate) to raise runoff coefficient, Microcatchment (negarim), contour bunds, contour furrows for trees.
* Pond capacity: V = (catchment area x rainfall x runoff coeff.) - losses; design to hold 1-2 irrigations.
* Types by scale: micro-catchment (in situ), macro-catchment (50 - 1000 sq m catchment for one field), floodwater harvesting (spate irrigation), roof and ground catchment.

6. Harvesting Structures

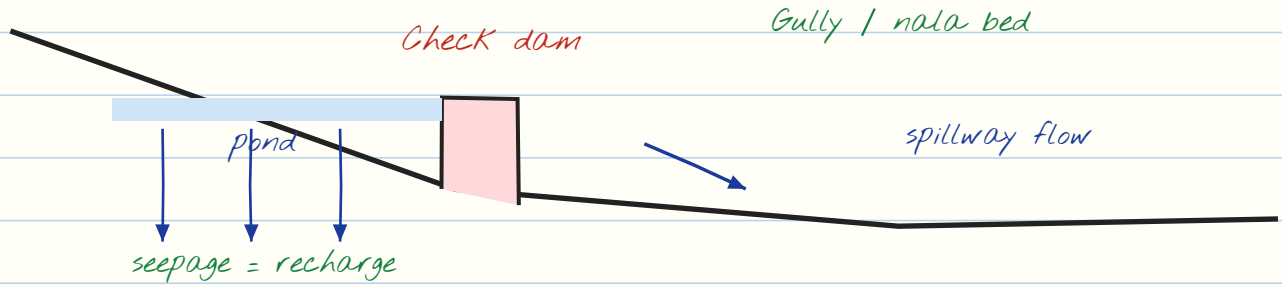


Fig 2: Check dam with pond and recharge

Structure	Features / use
Farm pond	dugout or embankment 0.5–3 m depth; protective irrigation, fish; lined to cut seepage
Contour bund / trench	on slopes 2–6 %; hold runoff, increase infiltration; spacing by slope
Gully plug / loose boulder	small structures across gullies; trap silt, use local stone
Check dam (masonry/earthen/gabion)	across nala; 0.5–3 m high; pond + recharge; spacing by slope
Nala bund	earthen bund across a nala with spillway; 1 – 3 m; for streams with 3 – 10 % slope
Percolation tank	pond on permeable ground for groundwater recharge; downstream wells benefit
Sub-surface dyke (underground bandhara)	impervious wall below river bed to stop sub-surface flow in sandy beds
Khadin / johad / tanka	traditional Rajasthan structures; Khadin = bunded field, johad = earthen check dam, tanka = underground tank
Stop dam / weir, Kolhapur-type (KT) weir	barrage with gates to store river water

* Site selection: narrow section with good foundation, large catchment, stable banks, permeable strata downstream (for recharge), away from habitat.
* Design of check dam: height H, length L of spillway ($Q = 1.7\ L\ H^{1.5}$), apron and cut-off wall, wings, protection of toe from scour. Spacing: Vertical interval = 1 – 2 m, Spacing = H / slope.
* Maintenance: de-silting, repairs of damaged section, checking of leakage, community ownership and water user group.

Sequence:

Ridge-to-valley sequence: contour trenches on hills -> gully plugs -> check dams in drainage lines -> farm ponds and percolation tanks in lower reach.

7. Model Watershed

Model watershed: a demonstration watershed developed in a participatory manner with the best technology so that other villages can learn and replicate. It shows integrated soil, water, crops, livestock and institutions.

* Features: fully treated ridge to valley, community institutions (watershed committee, SHGs, user groups), self-sustained maintenance fund, monitoring of runoff and sediment at the outlet, crop diversification and income generation.
* Examples in India: Ralegan Siddhi (Maharashtra, Anna Hazare) - water conservation + social reforms; Hiware Bazar (Maharashtra, Popatrao Pawar) - 'model water-rich village'; Sukhomajri (Haryana) - silt control + community sharing of water; Ralegaon; Adgaon; Kothapally (Telangana - ICRISAT consortium); Rajasthan - Tarun Bharat Sangh (Rajendra Singh) revival of Arvari river by johads; Jhabua (MP) - Livelihood-based approach.
* ICAR model watersheds in 47 centres; NRAA and CRIDA (Hyderabad) developed best practice; 'Watershed plus' approach.
* Success factors: leadership, participation, transparency, equity, gram sabha decision, social fencing (ban on grazing, ban on tree felling, ban on water-hungry crops), simple technology, capacity building, linkage to markets.

Village	Key actions	Result
Ralegan Siddhi	check dams, percolation tanks, afforestation, grazing ban, prohibition	water table up, double crops, migration stopped
Hiware Bazar	trenches, 'water budget' per year, crop planning	poverty fell, income per capita rose many times
Sukhomajri	Sukhna lake silt control by dams + water rights	sediment in lake reduced > 90 %
Arvari (Alwar)	300+ johads, community Arvari Sansad	river flows perennially again

8. Government and NGO Projects

Government programmes

* DPAP (Drought Prone Areas Programme, 1973), DDP (Desert Development Programme, 1977), IWDP (Integrated Wastelands Development Programme, 1989).
* NWDPRA (National Watershed Development Project for Rainfed Areas, 1990-91), River Valley Project (RVP) and Flood Prone River (FPR) schemes for catchments of dams.
* Hariyali guidelines (2003): Panchayati Raj institutions as the planning and implementing agency.
* IWMP (2009-10) under Dept of Land Resources, then PMKSY - WDC (2015-16): Pradhan Mantri Krishi Sinchayee Yojana (Har Khet Ko Pani, 'more crop per drop'), WDC-PMKSY 2.0 (2021-26).
* MGNREGA - works on water conservation, water harvesting and drought-proofing; Jal Shakti Abhiyan - Catch the Rain; Atal Bhujal Yojana (community groundwater management in 7 states); Mission Amrit Sarovar (75 water bodies per district); National Mission for Clean Ganga, Namami Gange; National Wetland Conservation Programme (NPCA).

* Common Guidelines (2008): shared cost 90:10 (Centre:State), project cost Rs 12,000 - 15,000 per ha; 1 % M and E, 10 % administration, 5 % institution, 50 % works.

NGO and international projects

* BAIF (Pune), WOTR (Pune), Tarun Bharat Sangh (Rajasthan), Foundation for Ecological Security (FES), Sadguru Foundation (Gujarat), MYRADA (Karnataka), Aga Khan Rural Support Programme, DHAN Foundation (tanks), PRADAN, Watershed Support Services and Activities Network (WASSAN).
* Donor assisted: Indo-German Watershed Programme (Maharashtra), Indo-Swiss, World Bank-assisted Karnataka Watershed Development (Sujala), Andhra Pradesh Rural Livelihood Project, DFID, IFAD, ICRISAT 'Consortium approach', NABARD (WDF - Watershed Development Fund).
* Role of NGOs: facilitation, social mobilisation, training, technical support, monitoring, building community institutions, linking with government schemes.

Success:

Common elements of successful projects: people's participation, ridge-to-valley treatment, equity, capacity building, convergence of schemes, sustainability fund.